

**Kathmandu University School of Education
Department of STEAM Education**

**Bachelor of Technical Education (B Tech Ed) Program
Information Technology (IT)**

In Association with the School of Engineering, KU

Course Curriculum

Semester VII

2025

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Overview of the B Tech Ed Program

Technical and Vocational Education and Training (TVET) has been one of the prominent sectors globally for a long. Dewey (1916) saw TVET as a tool for education reform in modernizing society. This has been equally vibrant in the present context as scholars (e.g., Maclean & Wilson, 2009) argue that TVET is an education and training that prepares an individual for gainful employment. TVET programs equip young people with the skills, knowledge, and competencies required to enter a particular type of professional career (OECD 2017). Hence, the role of TVET in preparing skilled human resources and enabling them to transition into the career path for gainful employment in specific and the country's economic development in a broad is inevitable. The formal TVET began in Nepal only around mid of the 20th century. Nevertheless, occupational skills have been transformed from generation to generation for a long (Ministry of Education [MoE], 2012). The establishment of the Council for Technical Education and Vocational Training (CTEVT), the governing body of TVET in 1989, was a substantive effort toward its development and expansion (CTEVT, 2019). Additionally, at present, the Centre for Education and Human Resource Development (CEHRD) offering technical diplomas in 284 technical stream schools throughout the country (GoN, 2018) and universities (such as Kathmandu University, Manmohan Technical University, etc.) run bachelor and master in TVET and other 11 federal ministries offer TVET related formal and non-formal programs (MoEST, 2019). The CTEVT, which is in charge of formulating TVET-relevant policies, developing standards for programs and curriculums, coordination, accreditation, monitoring, and supervision (CTEVT, 2019) largely shares the TVET related activities. There are 1131 CTEVT affiliated and constituent institutions with roughly the average capacity of 70,000 per annual operate formal TVET programs (CTEVT, 2020). Further expansion of TVET institutions in the country is yet to be expected soon since the government has a provision of establishing at least one technical school in all local levels.

Bachelor of Technical Education is a program to develop instructors and teachers in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the 'practical focus' of the program, the assessment has been proposed to address the needs of experiential learning.

The 136-credit program has been structured four types of the courses, such as Educational Core (18), Vocational Specialization (80), Vocational Pedagogy (20), and Educational Research and Internship (18) with a view to inculcate both pedagogical and vocational skills. The vocational specializations have been offered in a manner that they can be utilize hands-on activities as well. The program begins with the foundational courses in general education, vocational pedagogy and the area of specialization. In the later part of the program, OJT/WBL and forms of experiential learning shall be introduced. The culmination of the program will be internships in their specializations and a research report that addresses the problem of vocational education.

B Tech Ed in Information Technology

Bachelor of Technical Education is a 4-year (8 semesters) program to develop instructors, teachers, professionals in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the ‘practical focus’ of the program, the assessment has been proposed to address the needs of experiential learning. The B Tech Ed in IT program aims at providing technical, vocational, and educational exposures for students who can become IT professionals, IT Instructors, TVET educators, TVET curriculum developers, etc. The occupational/specialization courses emphasize the development of core IT skills in the field of computer science and technology innovation. The B Tech Ed in IT graduates will be eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Major Objectives of the B Tech Ed in IT Program

- Demonstrate Level 6 competencies envisaged by the National Vocational Qualification Framework (NVQF) of Nepal,
- Exhibit comprehensive, meaningful, and coherent knowledge and skills in the field of technology and computer science
- Apply occupational specialization knowledge and skills in solving problems occurred in their contexts of work,
- Apply vocational pedagogical knowledge and skills in technical and vocational instruction of Information Technology,
- Show scholarly literacy, communication, quantitative reasoning, critical thinking, learning skills needed for advanced and work-based learning,
- Exemplify a deep and principled understanding of the technical and vocational learning processes and the role of the instructors in facilitating these processes in the students,
- Apply a wide range of teaching process skills including curriculum development, lesson planning, materials development, assessment, and pedagogical approaches, and
- Enable students to become resilient and lifelong learners by harnessing internal and external assets as TVET professionals.

Career Prospects

- TVET Instructors and Educators
- TVET Curriculum Developers
- IT Professionals (Software and Apps developer, programmer, Network Engineer, Web Developer, IT Admin, IT Engineer, etc.)
- Eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

Entry Requirements

The minimum prerequisite for this program is the completion of Level 5 education (as per National Vocational Qualification Framework) in relevant vocational specializations. The details are-

- Minimum 1.6 CGPA in 10+2
- Or
- A minimum 40% marks in 10+2, PCL or equivalent in any disciplines

Courses Offered in the Semester V

Subject		Credit Hour		
		Theory	Practical	Total
ITEX 410 Big Data		2	1	3
ITEX 400 E-Governance		1	1	2
VPED 441 Internship – Teaching/Training		0	6	6
ITEX 499 Professional IT Project- II		0	3	3
	Total	2	12	14

Research and Internship

VPED 441 Internship – Teaching/Training

Course Name: Internship - Teaching Course Curriculum

Code: VPED 441

Credit Hours: 6 Credit Hours (90 hours of learning engagement)

Nature of the Course: Practice-based, Practical

Duration: 15 weeks (full-time internship, approximately 6 hours per week in the field, plus preparatory and reflective activities)

Course Description

The Internship - Teaching Course Curriculum provides 4th-year students in the Bachelor of Technical Education in Information Technology program with a structured opportunity to apply vocational pedagogical knowledge and IT expertise in real-world classroom settings. The course emphasizes hands-on teaching experience, resource development, and reflective practice to prepare students for professional IT education roles.

Students will design and deliver IT lessons, develop tailored teaching resources (e.g., lesson plans, lab manuals, digital tutorials), and address classroom challenges in technical schools or vocational institutes. Under mentor supervision, students will deliver IT lessons, create curriculum-aligned resources, and compile a reflective digital portfolio showcasing their teaching journey. The course integrates Nepal's NVQF Level 6 and IT curriculum standards, fostering skills in lesson planning, classroom management, student engagement, and professional collaboration. Assessments focus on teaching performance, resource quality, reflective practice, and professionalism, ensuring students emerge as competent, reflective IT educators ready to address context-specific educational challenges.

Course Objectives

By the end of the course, students will be able to:

1. Apply vocational pedagogical knowledge to design and deliver effective IT lessons in a classroom setting.
2. Develop practical teaching resources (e.g., lesson plans, assessments) to support IT education.
3. Demonstrate professional teaching skills, including lesson planning, classroom management, and student assessment.
4. Collaborate with mentors and school stakeholders to enhance IT teaching practices.
5. Reflect critically on teaching experiences to improve instructional practice and foster lifelong learning.
6. Address context-specific challenges in IT education using pedagogical and IT expertise.

Learning Outcomes

Upon successful completion, students will:

1. Deliver IT lessons, demonstrating effective teaching strategies and IT subject mastery.
2. Develop and implement effective lesson plans for IT topics, tailored to the school's curriculum.
3. Develop and implement effective teaching/learning resources (e.g., lab manual, digital tutorial, or assessment tool) for IT classrooms.
4. Produce a reflective digital portfolio documenting teaching experiences, including lesson plans, resources, and self-evaluations.
5. Demonstrate professional communication and collaboration with mentors, teachers, and students.
6. Propose and implement a solution to one classroom challenge (e.g., student engagement, outdated materials).

Course Structure

The course is divided into three phases: **Pre-Internship Preparation (15 hours)**, **Field-Based Teaching (65 hours)**, and **Post-Internship Reflection and Assessment (10 hours)**. The 90 hours are distributed across 15 weeks, with approximately 4 hours per week in the classroom and 2 hours per week on preparation, reflection, or assignments.

Breakdown of Hours

- **Pre-Internship Preparation:** 15 hours (Weeks 1–2)
 - Orientation, lesson/resource planning, and pedagogical workshops.
- **Field-Based Teaching:** 65 hours (Weeks 3–15)
 - Teaching (40 hours), resource development (15 hours), observation and collaboration (10 hours).
- **Post-Internship Reflection and Assessment:** 10 hours (Weeks 16–18)
 - Portfolio compilation, presentations, and assessments.

Course Content and Weekly Plan

Weeks 1–2: Pre-Internship Preparation (15 hours)

Objective: Prepare students for effective teaching in IT classrooms.

- **Activities:**
 - **Orientation Workshop (4 hours):** Overview of internship expectations, teaching roles, and alignment with Nepal’s NVQF Level 6 and IT curriculum of Nepal. Guest lectures from experienced IT teachers/faculties.
 - **Pedagogical Training (5 hours):** Workshops on lesson planning, active learning strategies (e.g., project-based learning, think-pair-share), and classroom management for IT subjects.
 - **Resource Plan Development (5 hours):** Workshops on developing effective resources (Lesson plans, assessment plans, activities, etc. based on progressive educational frameworks)
 - **Site Orientation (1 hour):** Virtual or in-person visits to internship sites to meet mentors and review the school’s IT curriculum.
- **Assignments:**
 - Submit two draft lesson plans with clear objectives, activities, and assessments.
 - Submit two sets of resources for an IT subject
 - Write a 300-word statement of teaching goals for the internship.
- **Resources:**
 - Sample lesson plans and resource planning templates
 - Nepal’s IT curriculum for school or TVET level
 - Pedagogical guides (e.g., Bloom’s Taxonomy for IT instruction, Constructivism, Progressive Learning, Inquiry-based Learning, etc.).

Weeks 3–13: Field-Based Teaching (65 hours)

Objective: Develop resources and apply teaching skills in real-world IT classrooms.

- **Sites:** Technical schools, vocational institutes, or schools with IT programs
- **Activities:**
 - **Teaching (40 hours):**
 - Deliver at least 40 hours of IT lessons to students in regular class time.
 - Use varied teaching methods (e.g., hands-on labs, coding projects, interactive quizzes, discussion, inquiry, demonstration, presentation, etc.).
 - **Teaching Resource Development (20 hours):**
 - Create resource materials for each classroom session including lesson plans and students learning materials (SLMs)

- Collaborate with mentors to ensure the resource aligns with the school's IT curriculum.
- **Observation and Collaboration (5 hours):**
 - Observe experienced IT teachers (at least 2/3 sessions) to learn their classroom practices, identify needs and issues, explore solution strategies, and implement in the classroom.
 - Participate in faculty meetings or curriculum discussions to understand school needs.
- **Mentor Supervision:**
 - Weekly feedback sessions with an on-site mentor (schoolteacher) and periodic check-ins and feedback session with a course supervisor/facilitator.
 - Mentors evaluate lesson delivery, student engagement, and resource quality.
- **Assignments:**
 - Maintain a daily teaching log detailing lessons taught, challenges, and reflections.
 - Submit the finalized teaching resource and plans every week.
- **Resources:**
 - School IT labs with software, curriculum materials, etc.
 - Access to both physical and digital resource materials.
 - Reference materials from global teacher education programs

Weeks 14–15: Post-Internship Reflection and Assessment (10 hours)

Objective: Synthesize teaching experiences and demonstrate professional growth using Digital portfolio platform (e.g., Google Sites).

- **Activities:**
 - **Portfolio Compilation (6 hours):** Students compile a portfolio including:
 - Objectives of the Portfolio and internship program
 - Lesson and Activity Plans
 - Teaching/Learning Resource Materials (Images or other documents)
 - Teaching Log and Reflections
 - Feedback logs from both the mentor and the faculty
 - Multimedia (Photos and Videos) of teaching, student engagement, and feedback sessions
 - List of new knowledge, skills, experiences, and challenges
 - A 1,000-word reflective essay on their development as an IT educator.
 - **Presentation (2 hours):** Students present their teaching experiences to peers and faculty (15-minute presentation + 10-minute Q&A), highlighting key lessons and challenges.
 - **Final Assessment (2 hours):** Evaluation by university supervisor based on portfolio, presentation, and mentor/faculty feedback.
- **Assignments:**
 - Submit the digital portfolio.
 - Submit the hard copy of all the evidence
 - Prepare and deliver the final presentation.
- **Resources:**
 - Digital Portfolio guidelines
 - Presentation rubrics and templates

Assessment Methods

Assessment focuses on teaching performance, resource development, and reflective practice, problem solving, and attitude of the student.

Internal Assessment (75%)	External Assessment (25%)
• Pre-internship Preparation (5%): This will be evaluated by the course facilitator • Teaching Performance (35%): Based on quality/effectiveness lesson delivery, student engagement, assessment strategies, and classroom management ○ On-site mentor Evaluation – 25% ○ Course Facilitator Evaluation – 10% -- Interaction, feedback, improvement, implementation of the • Teaching Resource (20%): This will be assessed by the facilitator based on quality/effectiveness of teaching/learning resource materials; relevance, practicality, curriculum alignment; availability; etc. • Teaching Log (10%): This will be evaluated by the field mentor (teacher) • Professionalism (5%): This will be assessed by mentor and supervisor for punctuality, collaboration, and communication.	• Digital Portfolio (15%): Based on the contents and presentation of the portfolio requirements • Final Presentation (5%): Based on the sharing and demonstration of the teaching/learning experience • Viva Voce (5%): Oral examination to discuss teaching experiences and challenges.

Summary of Assessment Components

Component	Evaluator	Percentage	Points
Pre-Internship Preparation	Course Facilitator	5%	5
Teaching Performance	On-Site Mentor (25%) + Course Facilitator (10%)	35%	35
Teaching Resource	Course Facilitator	20%	20
Teaching Log	On-Site Mentor	10%	10
Professionalism	On-Site Mentor + Course Facilitator	5%	5
Digital Portfolio	External	15%	15
Final Presentation	External	5%	5
Viva Voce	External	5%	5
Total		100%	100

Professional IT Courses

ITEX 410 Big Data

Course Title: Big Data

Credit Hours: 3

Course Code: ITEX 410

Nature: Theory and Practical

Course Description and Goals

This course will provide students with a broad introduction to Big Data technologies, including Hadoop-based architecture, data ingestion, data transformation, data management, analytics, and predictive analytics for manipulating and discovering insights. Topics include Hadoop, HDFS, MapReduce, Spark, Hive, and Pig. The course will have significant hands-on sessions and draw from case studies and applications.

Learning Outcomes

On completion of this course, the students will be able to:

- Explore a sound understanding of the principles of organization, validation, and transformation, and analyze large volumes of data on specialized platforms (Big Data) from various data sources: files, databases, server logs, etc.
- Use the Big Data platform ecosystem for processing Big Data
- Understand the potential use in a corporate environment
- Comprehend the advantages and limitations of Big Data technologies, including predictive analytics, and build the confidence to interpret data as insights to drive organizational success.

Content with specific objectives

Specific objectives	Contents
1. Understand big data concepts 2. Familiar with big data processing and analytics in education 3. Understand optimization and performance of big data ecosystem	Unit One: Introduction to Big Data (4 hours) 1. Big data: definition and taxonomy 2. Big data value for education 3. Handling and processing big data 4. Methodological challenges and problems in big data applications 5. Data summarization and visualization
1. Understand Hadoop's core components 2. Explore data ingestion & storage 3. Explore Hadoop trends 4. Understand data processing with Mapreduce, Pig and Oozie	Unit Two: The Hadoop Ecosystem (6 hours) 1. Introduction to Hadoop 2. Hadoop components: MapReduce/Pig/Hive/Hbase/Yarn/Spark/Storm 3. Loading data into Hadoop 4. Handling files in Hadoop 5. Getting data from Hadoop 6. HDFS 7. Hadoop YARN Architecture 8. Exploring Pig and Oozie

1. Learn Hive architecture 2. Explore data management in Hive 3. Build Hive-based ETL pipelines 4. Fix common Hive errors	Unit Three: Querying Big Data with Hive (5 hours) 1. Introduction to the SQL Language 2. From SQL to HiveQL 3. Using Hive to query Hadoop files 4. Hive for massively parallel on disk data processing
1. Understanding Data Ingestion Fundamentals 2. Mastering Apache Flume for Log Aggregation 3. Learning Kafka for Scalable Event Streaming	Unit Four: Data Ingestion using Apache Flume and Kafka (4 hours) 1. An overview of batch and real-time data ingestion patterns using Apache Flume and Kafka 2. Data transformation techniques and generation of summary statistics using Apache Spark
1. Scaling Machine learning with Big Data Technologies 2. Understand Big Data and Machine learning Use Cases 3. Deploy, monitor, and optimize Machine learning systems in big data	Unit Five: Big Data and Machine learning (5 hours) 1. Big Data & Machine Learning 2. Machine learning tools: Spark & SparkML, H2O, Azure ML 3. Big data analysis using graph databases: Analyzing big data with Neo4j using Cypher queries.
1. Understanding Graph-Based Data Representation 2. Mastering Graph Algorithms for Big Data 3. Explore Graph Databases and Query Optimization	Unit Six: Big Data Analysis using Graphs (6 hours) 1. Introduction to graphs 2. Graph analytics: Path analytics, connectivity analytics, community analytics, centrality analytics 3. Graph analytics with Neo4j, a graph database, cypher query language 4. Graph analytics with GraphX, Apache Spark's API for graphs and graph- parallel computation

Major Teaching and Learning Strategy (List of some Teaching and Learning Strategies for the faculty)

- **Blend theory and hands-on labs**
- **Use cloud platforms and sandboxes**
- **Introduce Industry-aligned projects and certifications**
- **Faculty Development Programs** on Hadoop, Spark and Cloud
- **Assign real-world datasets** (e.g., Twitter feeds, IoT sensor logs)
- **Ethical considerations of big data and modern Education Technology tools**

Practical (List some of the practical and field-based tasks to be done)

- Combine lectures on Big Data concepts (distributed computing, Hadoop/Spark, NoSQL) with hands-on labs.
- Work with Hadoop, Spark, Kafka, and cloud platforms (AWS, GCP, Azure).
- Build end-to-end Big Data pipelines (e.g., X (Former Twitter) sentiment analysis, log analytics).

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
LAQ	16 Marks	3 Questions [Attempt 2 Questions]

Marking Scheme for Final Examination:

- Section A (**MCQ**): 10 Questions $\times$ 1 marks
- Section B (**Short Answer Question-SAQ**) – Total 7 Questions $\times$ 4 marks (Attempt any Six)
- Section C (**Long Answer Question-LAQ**) – Total 3 Questions $\times$ 8 marks (Attempt any Two)

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Introduction of Big Data	4 hours	1	1	-	
The Hadoop ecosystem	6 hours	3	2	1	
Querying big data with Hive	5 hours	1	1	-	
Data Ingestion using Apache Flume and Kafka	4 hours	2	1	-	
Big data & Machine learning	5 hours	1	1	1	
Big data analysis using graphs	6 hours	2	1	1	

References:

1. White, T., (2015) Hadoop: The definite Guide. Sebastopol: O'Reilly & Associates. ISBN10: 1491901632 [CORE].
2. Hadoop For Dummies, Dirk deRoos, For Dummies, 2014.
3. Viktor Mayer-Schönberger & Kenneth Cukier (2013). **Big Data: A Revolution That Will Transform How We Live, Work, and Think.**
4. Nathan Marz & James Warren (2015). **Big Data: Principles and Best Practices of Scalable Real-Time Data Systems.**

ITEX 400 E-Governance

Course Title: E-Governance

Course Code: ITEX 400

Credit Hours: 2 (1+1)

Nature: Theory + Practical

Course Description and Goals

This course introduces students to the principles and practices of E-Governance with a focused lens on Nepal's education system. It explores how digital technologies, public policy, and governance frameworks come together to improve transparency, service delivery, and decision-making in schools, colleges and universities. Students will examine real-world initiatives from Nepal and beyond, including digital education platforms, government portals and data-driven planning tools. Through interactive sessions, local case studies and critical reflection, learners will build the skills and perspective needed to actively contribute to Nepal's digital transformation in education.

Learning Outcomes

On completion of this course, the students will be able to:

- Differentiate between E-Government and E-Governance, particularly in the context of education.
- Analyze models of digital governance and understand how public-private partnerships (PPP) support service delivery in education.
- Assess the digital readiness of Nepal's education sector using recognized global frameworks.
- Understand how infrastructure, data systems and policy work together to shape inclusive and accountable governance.
- Reflect on national and international case studies to identify lessons that can inform meaningful change in Nepal's digital education landscape.

Course Content with Specific Objectives

Specific objectives	Contents
• To understand what E-Government and E-Governance mean, and how they've evolved. • To clearly distinguish between the two concepts. • To explore how E-Government functions as an information system. • To recognize its key benefits, limitations, and common misconceptions. • To learn about different types of digital government services (G2C,	Unit 1: Introduction to E-Government and E-Governance (5 Hours) • Definitions, scope and evolution • Difference between E-Government and E-Governance • E-Government as an information system • Benefits, myths and limitations of E-Government • Lifecycle of E-Government and Types of E-Government services (G2C, G2B, G2G, G2E) • Current global trends in digital transformation of education systems

G2B, G2G, G2E) and the E-Government lifecycle. To get an overview of how education systems worldwide are being transformed by digital governance.	
- To explore key models of E-Governance and how they guide digital service delivery. - To understand the Interactive Service Model (G2C2G) and its practical use. - To learn about maturity and adoption stages in E-Governance development. - To grasp the concept of Public-Private Partnerships (PPP) in driving e-Government projects. - To examine real examples of how PPPs are helping shape digital education services in Nepal.	Unit 2: Models of E-Governance and PPP (7 Hours) - Broadcasting / Wilder Dissemination, Critical Flow, Comparative Analysis and Mobilization & Lobbying Model - Interactive Service Model (G2C2G) - Maturity and Adoption models - Public-Private Partnerships (PPP) in E-Government - PPP case applications in Nepal's digital education initiatives
- To identify the essential ICT infrastructure required in schools, colleges, and universities for effective e-Governance. - To understand how cloud computing and interoperability frameworks support connected, digital classrooms. - To explore how platforms like EMIS follow national and international standards to guide planning, monitoring, and service delivery. - To recognize the real-world importance of accessible, secure, and scalable digital systems in education.	Unit 3: Infrastructure and Standards for E-Government (5 Hours) - Network, computing and data center infrastructure in schools, colleges and universities - Interoperability frameworks and cloud-based tools to support digital classrooms - Education Management Information Systems (EMIS) and their integration in planning and monitoring - National and international standards for managing and safeguarding educational data (e.g., UNESCO ICT indicators)
- To understand how E-Government readiness is assessed using global indicators. - To identify common risks in education governance and explore ways to address them. - To get a basic understanding of how digital education projects are planned and rolled out.	Unit 4: Readiness, Risk Management, and Implementation (5 Hours) - E-Government readiness frameworks and indicators (UN EGDI, ITU ICT Index, GTMI, UNESCO ICT in Education) - Risk assessment and mitigation strategies in educational governance

To learn the role of SLAs, monitoring, and evaluation in ensuring accountability and performance.	- Basic project lifecycle stages for digital governance implementation - Introduction to Service Level Agreements (SLAs), monitoring, and evaluation practices
- To recognize common cybersecurity concerns in digital education platforms. - To gain a basic understanding of Nepal's cyber laws and global legal trends. - To reflect on the ethical use of data and the importance of student privacy.	Unit 5: Security and Legal Issues in Digital Governance (2 Hours) - Common security challenges in educational digital systems - Overview of Nepal's cyber laws and international best practices - Key issues in data privacy, ethical use and safe analytics in education
- To explore Nepal's major E-Governance initiatives in education and their local impacts. - To analyze India's federal approach to digital education infrastructure. - To compare global case studies and extract lessons relevant to Nepal. - To build a benchmark-based understanding of what effective digital education governance looks like.	Unit 6: Case Studies and Benchmarking (6 Hours) Nepal: Digital Nepal Framework (Education and Governance Pillars), EMIS and integration with Nagarik App, Teacher Service Commission (TSC) Portal, HRM systems and CTEVT's digital transition, Local-level implementations – Smart Palikas, GIDC and education dashboards India: DIKSHA platform and its ecosystem and National Digital Education Architecture (NDEAR) Global: Estonia's e-School and X-Road interoperability model, South Korea's Smart Campus strategy and USA's Open Data for education and accountability

Major Teaching and Learning Strategies

- Interactive Lectures:* Blend concepts with real-life digital tools and examples from Nepal's education sector to make sessions relatable and engaging.
- Case-Based Discussions:* Use case studies from Nepal, India, and global contexts to connect theory with lived experience and policy practice.
- Collaborative Activities:* Encourage small group tasks or discussion-based mini projects that explore challenges and solutions in educational governance.
- Practical Demonstrations:* Introduce tools like EMIS, Nagarik App, or open data dashboards in class to help students visualize real-world systems.
- Guest Sessions:* Invite experts from MoEST, MoCIT, municipalities, or E-Governance projects to share practical insights.

- *Reflective Assignments:* Ask students to analyze how digital governance affects their own learning environments, communities, or institutions.

Practical and Field-Based Tasks

- *Platform Review:* Explore and critique platforms like EMIS, Nagarik App, or IRD with a focus on usability, accessibility, and relevance to education users.
- *Mini Proposal Exercise:* Develop a short concept for improving a specific aspect of digital governance in a school or college (e.g., attendance tracking, student feedback system).
- *Data Interpretation:* Work with publicly available education datasets to create insights for planning or resource distribution.
- *Policy Brief:* Draft a short brief recommending improvements to Nepal's digital education governance, referencing global benchmarks.
- *Field Interview:* Engage with local education officers, school administrators, or ICT coordinators from colleges and universities to understand challenges and gaps in implementing digital governance systems.

Assessment Plan

- **In-semester (50)**
Assignment, written exam, presentation, case study, report, objectives, attendance
- **End-semester (50)**

Question Types	Marks Distribution	Total Question
MCQ	10 Marks	10 Questions
SAQ	24 Marks	7 Questions [Attempt 6 Questions]
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Marking Scheme for Final Examination:

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- Section C (**Long Answer Question-LAQ**) – Total 3 Questions × 8 marks (Attempt any Two)

Note: While this is a theory-based course (2 credits, 30 lecture hours), students are encouraged to engage in field observations, digital platform reviews and policy reflection activities. These applied tasks are designed to enhance understanding of how E-Governance works in real educational settings and may be incorporated into internal assessments where appropriate.

Learning Units	Total Credit Hours	MCQs	SQs	LQs	Remarks
Unit 1: Introduction to E-Government and E-Governance	5	1	1	0	

Unit 2: Models of E-Governance and PPP	7	2	2	0	
Unit 3: Infrastructure and Standards for E-Government	5	2	1	1	
Unit 4: Readiness, Risk Management, and Implementation	5	2	1	1	
Unit 5: Security and Legal Issues in Digital Governance	2	2	1	0	
Unit 6: Case Studies and Benchmarking	6	1	1	1	
Total	30	10	7	3	

References and Further Reading

Core Textbooks

- Prabhu, C. S. R. (2012). *E-governance: Concepts and case studies*. PHI Learning.
- Heeks, R. (2006). *Implementing and managing e-government*. Sage Publications.
- Satyanarayana, J. (2006). *E-government: The science of the possible*. Prentice Hall India.
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- Anthopoulos, L. (2017). *Understanding smart cities: A tool for smart government or an industrial trick?* Springer.

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- Backus, M. (2001). *E-governance in developing countries*. IICD Research Brief No. 1.
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- OECD. (2020). *Digital government index and governance for youth-centric digital services*.
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ITEX 499 Professional IT Project – II

Course Title: Professional IT Project – II

Credit Hours: 3 (Practical)

Course Code: ITEX 499

Nature: Practical

Project Description and Goals

Professional IT Project II is a rigorous, project-based course rooted in the philosophy of learning through purposeful creation. The course is structured around **individual project execution**, ensuring that **each student independently** navigates the full spectrum of professional software development—from conceptualization to deployment—with guidance from a faculty supervisor.

This is not a group assignment; **each student must independently demonstrate their technical competency**, educational insight, and creative problem-solving ability. The **project must strictly align with the education domain**, reflecting the program’s core mission to empower learners who can innovate at the intersection of technology and pedagogy.

Students are encouraged to select challenges grounded in the **real needs of classrooms, educational institutions, educators, or learners**. Projects may take the form of **web-based, desktop, or mobile applications**, depending on the context and solution design. The focus should remain on solving authentic problems through technically sound and educationally relevant innovations.

Examples include (but are not limited to):

- Designing a **web-based learning analytics dashboard** to help teachers visualize student performance trends and identify at-risk learners.
- Creating a **mobile or desktop application** for inclusive learning tailored to students with low vision, dyslexia, or other learning differences.
- Building an **exam seating automation system** for universities that ensures fairness and logistical efficiency, with options for both desktop and web deployment.
- Constructing a **digital attendance system** that integrates QR code scanning or facial recognition, suitable for both web portals and school computer labs.

Each student will be mentored by a designated supervisor who provides academic and technical guidance throughout the project. At the culmination of the course, **students will present their work to a panel that includes external examiner**. The final evaluation will be based on the originality, technical soundness, usability, educational relevance, and professional presentation of the project. To ensure authenticity and accountability, students are required to maintain **individual log sheets** that document weekly progress, technical decisions, reflections, and evolving challenges. This log becomes a meta-narrative of their learning journey and plays a vital role in the overall assessment.

Ultimately, this course is more than a capstone—it is a proving ground. It is where students internalize the professional ethos of IT development, not as coders alone, but as thoughtful innovators capable of contributing to a smarter, more inclusive educational future.

Learning Outcomes

On completion of this course, the students will be able to:

- Independently identify a meaningful educational problem and translate it into a clearly defined IT-based project scope grounded in real-world needs.
- Design and develop a functional IT solution—such as a web-based, desktop, or mobile application—that addresses specific challenges faced by learners, educators, or institutions.
- Apply structured software development methodologies, including system analysis, design, implementation, and testing, in a way that reflects sound engineering practice.
- Integrate educational context and pedagogical relevance into technical design decisions, ensuring that solutions are not only functional but also meaningful and usable in learning environments.
- Document the software development process professionally through comprehensive technical reports, diagrams, logs, and reflective writing aligned with academic and industry standards.
- Demonstrate ethical integrity and professional responsibility, including proper attribution of code and ideas, accessibility considerations, and data privacy compliance.
- Present, defend, and justify the developed system through a structured demonstration and viva session, articulating both technical depth and educational impact.
- Reflect critically on personal learning, technical growth, and project challenges, identifying areas for future development and continuous improvement as a technology educator or practitioner.

Phases of the Project

The Professional IT Project II is organized into three progressive phases, each designed to scaffold student development from ideation to demonstration. Students are expected to maintain consistent weekly progress, engage in reflective documentation, and respond to supervisory feedback with professionalism and discipline. The **entire project spans 15 weeks**, culminating in a **Final Defense during Week 16**, where external examiner will assess each student's work.

Phase 1: Proposal Submission and Defense (Week 1–3)

This phase focuses on defining the problem and laying the groundwork for the solution.

- Students begin by identifying a relevant educational challenge and drafting a clear, feasible project proposal.
- The proposal should include: title, problem statement, objectives, scope, literature review, methodology, proposed tools, and a preliminary Gantt chart.
- By Week 3, each student must present and defend their proposal before an internal review committee.
- Feedback from the proposal defense must be thoughtfully incorporated into the development plan moving forward.

Phase 2: Project Development and Mid-Term Review (Week 4–12)

This is the core phase of the course, emphasizing iterative development and continuous feedback.

- Students begin implementation in Week 4 and are expected to meet regularly with their assigned supervisor for feedback, troubleshooting, and refinement.
- By Week 12, each student must submit a structured progress report and deliver a Mid-Term Demonstration of the system's current functionality.
- The mid-term review includes:
 - Discussion of implemented features
 - System architecture and design artifacts
 - Testing and debugging status
 - Challenges encountered and resolved
- This checkpoint ensures that students are on track for successful completion and allows for early intervention where necessary.

Phase 3: Final Report Submission and Project Defense (Week 13–15)

This phase emphasizes refinement, documentation, and final readiness.

- Students must polish their systems and prepare a comprehensive final project report, including all required documentation, visuals, appendices, and reflections.
- The final report must adhere to prescribed academic formatting standards and be submitted to the department by Week 15.
- A soft copy must also be shared with the external examiner in advance to facilitate review.

Final Defense and Demonstration (Week 16)

The course culminates in a formal Final Defense.

- Each student will present their completed system, justify design decisions, and demonstrate core functionalities.
- The defense includes:
 - Live demonstration of the software
 - Technical and reflective presentation
 - Viva-voce with internal and **external examiner appointed by the university**
- Evaluation will consider not only the final product but also the student's depth of understanding, ability to respond to inquiry, documentation quality, and individual contribution.

Assessment Plan

To ensure fairness, transparency, and academic integrity, the evaluation is carried out in two major stages: **Supervisor Evaluation (75%)** and **External Evaluation (25%)**.

Supervisor Evaluation (75%) – Internal Assessment

Conducted continuously across the semester, this component assesses the student's progress, discipline, and individual commitment. Supervisors will use a mix of fortnightly observations, mid-term reviews, and final documentation to assess the following areas:

Sub-Category	Weight
Work Done (analysis, coding, UI, testing) • Quality of system design and implementation • Understanding of tools, methods, and models • Completion level and effectiveness of solution	60%
Documentation (project report + logbook quality) • Clarity, completeness, formatting, and relevance	15%
Total (Supervisor + Internal Examiner)	75%

Note: The supervisor will assess 60%, and the internal examiner (regular faculty) will contribute the remaining 15%.

External Evaluation (25%) – Final Defense

This stage is conducted during **Week 16** and involves an **external examiner** appointed by the university. It includes a formal project defense, where the student must present and justify their work through a structured:

- Live demonstration of the application/system
- Technical presentation with architecture, tools, and logic
- Viva voce to assess understanding, originality, and clarity

Sub-Category	Weight
Project Presentation & Demonstration	10%
Technical Understanding (Viva)	5%
Final Report & Educational Relevance	10%
Total (Evaluated by External Examiner)	25%

Minimum Requirement for Passing

- Students must **pass both internal and external** stages independently.
- Failing to secure the minimum passing criteria in either will result in an **overall failure**, regardless of cumulative marks.
- The **final evaluation committee** will include:
 - Project Supervisor
 - Program Coordinator / HOD
 - Internal Examiner (Faculty Member)
 - External Examiner (Appointed by Kathmandu University)

Professional IT Project II – Project Proposal Format

A. Preliminary Section

- Cover Page
 - University Name
 - Department
 - Course Title: ITEX 499 Professional IT Project - II
 - Project Title
 - Student Name
 - Registration No.
 - Date of Submission

B. Body Section

1. Introduction
 - Provide a brief overview of the project and its relevance to the education domain.
2. Problem Statement
 - Clearly define the core issue or gap that your project aims to solve. Justify its educational significance.
3. Objectives
 - List the specific goals the system is expected to achieve (use bullet points for clarity).
4. Methodology
 - 4.1 Requirement Identification
 - What data or functional needs have been identified?
 - How was this need verified (user interviews, observation, etc.)?
 - 4.2 Feasibility Study
 - Technical Feasibility: Tools, frameworks, platforms available

- Operational Feasibility: Ease of use, relevance in the education sector
- Economic Feasibility: Free/open-source stack, personal effort only

4.3 Literature Review / Related Work

- Briefly review similar projects or systems
- What lessons can you learn from them?

4.4 Proposed System Design Approach

- State whether you're using a Structured or Object-Oriented methodology
- Mention tools for modelling (DFD, UML, etc.)

5. Tools and Technologies

- List the major tools, languages, libraries, and platforms you plan to use.

6. Expected Outcome

- Describe the deliverables—what will be built, and what impact it is expected to make in the educational domain.

7. Project Schedule (Gantt Chart)

Attach or embed a simple weekly timeline showing milestones for:

- Proposal Defense
- Requirement Analysis
- Design
- Implementation
- Testing
- Final Documentation
- Defense Preparation

8. References

- List sources referred to while preparing the proposal (use APA 7th format).

9. Optional (if available at proposal stage)

- Mockups/Wireframes
- Preliminary ER Diagram or Use Case Diagram

Professional IT Project II – Final Report Content Structure

A. Preliminary Section

1. Cover Page
 - University, Department, Project Title, Student Name, Reg. No., Place, Date
2. Student Declaration
 - Statement of originality and authenticity
2. Certificate by Supervisor
 - Official endorsement of the project work
3. Approval Sheet
 - Signatures from supervisor, coordinator, internal and external examiner
4. Acknowledgements
 - Acknowledging individuals and institutions who supported the project
5. Abstract
 - A brief summary (150–250 words) covering the project's background, objectives, methodology, tools used, and expected or achieved outcomes. Should provide a standalone overview of the entire project.
 - Keywords: 4–7 terms representing core themes, technologies, and application domain.
6. Table of Contents
7. List of Abbreviations
8. List of Figures
9. List of Tables

B. Main Report

Chapter 1: Introduction

- 1.1 Introduction
 - Context and overview of the project domain (education-focused)
- 1.2 Problem Statement
 - Clear articulation of the challenge or gap addressed
- 1.3 Objectives
 - Specific, measurable goals of the system
- 1.4 Scope and Limitations
 - What the project covers—and what it does not
- 1.5 Development Methodology
 - Overview of the adopted software development approach (e.g., Agile, Waterfall)
- 1.6 Report Organization
 - Brief description of the structure of the document

Chapter 2: Background and Literature Review

2.1 Background Study

- Concepts, technologies, and educational theories relevant to the problem domain

2.2 Review of Related Work

- Existing systems, research papers, or educational platforms addressing similar challenges

2.3 Critical Gap Identification

- What existing solutions miss and how your project fills that gap

Chapter 3: System Analysis and Design

3.1 System Analysis

3.1.1 Requirement Analysis

- Functional Requirements (with Use Case Diagrams/Descriptions)
- Non-functional Requirements (security, usability, performance)

3.1.2 Feasibility Study

- Technical Feasibility
- Operational Feasibility
- Economic Feasibility

3.1.3 System Modelling (Choose One Approach Below):

Structured Approach:

- ER Diagram
- Data Flow Diagrams (DFD – Context + Level 1 + Level 2)

Object-Oriented Approach:

- Class Diagram
- Object Diagram
- Sequence Diagram
- State Transition Diagram
- Activity Diagram

3.2 System Design

3.2.1 Architecture Design (Client-server, layered, microservices, etc.)

3.2.2 Database Schema Design (Tables, keys, relationships)

3.2.3 Interface Design (UI/UX mockups or wireframes)

3.2.4 Component/Deployment Diagrams (for OO approach)

3.3 Algorithmic Details (If Any)

- Description of core algorithms, logic, or decision-making models used in the system

Chapter 4: Implementation and Testing

4.1 Implementation

4.1.1 Tools and Technologies Used

- Languages, frameworks, libraries, CASE tools, platforms

4.1.2 Module-wise Implementation Description

- Describe key functions, classes, methods, or APIs used

4.2 Testing

4.2.1 Unit Testing

- Test cases, coverage, screenshots or outputs

4.2.2 System Testing

- Integrated workflow validation and bug fixing reports

4.2.3 User Acceptance Testing (if applicable)

- Summary of feedback from sample users/stakeholders

Chapter 5: Conclusion and Recommendations

5.1 Conclusion

- Summarize achievements relative to objectives

5.2 Lessons Learned

- Technical and personal growth insights during the project

5.3 Future Enhancements

- Scalability, usability, or performance features to add in future

Supplementary Sections

- References
 - APA 7th Edition formatting only for all cited sources
- Appendices
 - Screenshots of the system
 - Code snippets or GitHub repository link
 - **Supervisor log sheet (signed)**
 - Gantt Chart
 - Survey forms or interview transcripts (if used)

Project Report Formatting Guidelines

To ensure consistency, professionalism, and academic integrity, all submitted project reports must adhere to the following formatting standards. These guidelines are essential for readability, archival purposes, and presentation to internal and external examiner.

1. Page Numbering

- Pages from the Student Declaration up to (but not including) Chapter 1 must use lowercase Roman numerals (ii, iii, iv, ...).
- Pages starting from Chapter 1: Introduction onward must use Arabic numerals (1, 2, 3, ...).
- All page numbers should be centered at the bottom of the page.
- Each new chapter must begin on a new page.

2. Paper Size and Margins

- Paper Size: A4
- Margins:
 - Left: 1.5 inch (for binding)
 - Top, Bottom, Right: 1 inch

3. Font and Paragraph Styling

- Font Type: Times New Roman
- Font Size: 12 pt for body text
- Line Spacing: 1.5 lines throughout (except block quotations, which may be single-spaced)
- Paragraphs:
 - Fully justified
 - Use a first-line indent of 0.5 inch
 - Leave no extra spacing between paragraphs

4. Heading Styles and Hierarchy

To maintain APA 7 compliance, use a maximum of **five heading levels**, but typically:

Heading Level	Font Size	Style	Alignment
Chapter Titles	16 pt	Bold	Centered
Section Headings	14 pt	Bold	Left-aligned
Sub-section Headings	12 pt	Bold	Left-aligned
Sub-sub-section	12 pt	Italicized	Left-aligned
Inline Heading	12 pt	Bold Italic	Inline

5. Figures and Tables

- Placement: Center-align both figures and tables.
- Captions:
 - Figures: Caption should be centered below the figure.
 - Tables: Caption should be centered above the table.
- Caption Format: Times New Roman, 12 pt
- Numbering: Use chapter-based numbering.
 - Example: **Figure 2.1:** UI Mockup of Student Dashboard

6. Referencing and Citation – APA 7th Edition

- All in-text citations and reference listings must follow the APA 7th Edition style.
- Examples:
 - *(Author, Year)* for in-text
 - Lastname, F. M. (Year). *Title of the book*. Publisher.
- Use hanging indent (0.5 inch) for the reference list.
- List only the sources cited within the report in the References section.

7. Printing and Binding Recommendations

- Printing: Preferably single-sided.
- Binding Style: Hard with black cover and golden embossed text.
- Copies Required: At least 3 (Student Copy, Department Copy, and Dean's Office Copy)

(A typical Specimen of Cover Page / Title Page)

TITLE OF THE PROJECT

BY

Student's Full Name

Student's K.U. Registration No.

College Roll No.

Name of the KU School/Campus/College

A Project Report Submitted to

School of Education, Kathmandu University

in partial fulfillment of the requirements for the degree of
Bachelor of Technical Education in Information Technology

Place

Month, Year

STUDENT DECLARATION

This is to certify that I have completed the project entitled **Title of the Project** under the guidance of **Name of the Supervisor** in partial fulfillment of the requirements for the degree of **Bachelor of Technical Education in Information Technology** at the School of Education, Kathmandu University.

This is my original work and I have not submitted it earlier elsewhere.

Signature

Name of the Student:

K.U. Registration Number:

Date:

CERTIFICATE FROM THE SUPERVISOR

This is to certify that the project entitled **Title of the Project** is an academic work done by **Student Name** submitted in partial fulfillment of the requirements for the degree of **Bachelor of Technical Education in Information Technology** at the School of Education, Kathmandu University under my guidance and supervision.

To the best of my knowledge, the information presented by him/her in the project report has not been submitted earlier.

Signature of the Supervisor

Name:

Designation:

Date:

APPROVAL SHEET

This is to certify that the project titled **Title of the Project** submitted by **Student Name** has been examined and approved. In our opinion, it meets the required scope and quality standards for a project submitted in partial fulfillment of the requirements for the degree of **Bachelor of Technical Education in Information Technology** at the School of Education, Kathmandu University.

S.No.	Name	Academic Designation	Signature
1.	Project Supervisor		
2.	Program Coordinator / HOD		
3.	Internal Examiner		
4.	External Examiner		

Date of Defense:

Department: [Department Name]

Faculty: School of Education